Realtime Spatial-Temporal Graph Convolutional Network: Adapting for Continual Inference at the Edge

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***Abstract*—Spatial-Temporal Graph Convolutional Network (ST-GCN) is a well-established graph-based machine learning method for human action recognition, which processes temporal skeleton data to derive action predictions. Despite its novelty and advantages of graph processing, its current computational complexity and offline nature makes it challenging to deploy in continual action recognition use cases that need to run on resource-constrained embedded devices. In this work we propose a variant optimization, termed RT-ST-GCN, which adapts ST-GCN to realtime use cases without conceptually altering its architecture and to facilitate its deployment on wearable devices. On select hyperparameter ranges, our models show a reduction of 20-120x in complexity, 6-23x in cumulative memory, and 10-150x on-device acceleration. We compare our lightweight proposals against the original ST-GCN model on the PKU-MMD dataset, and show that they exhibit similar or better performance across key segmental indicators in spite of the decrease in model complexity. Lastly, we provide and benchmark a proof-of-concept implementation on a Raspberry Pi 4 to showcase the performance gains of RT-ST-GCN when running on a representative embedded platform. All our PyTorch codes are available at https://github.com/maximyudayev/Realtime-ST-GCN.**

***Index Terms*—Continual recognition, edge computing, graph convolution network, online action detection, realtime processing.**

I. Introduction

HUMAN action recognition (HAR) is an active field of research across a diverse set of domains [1]. It involves automatic identification of the most likely human action from a spatio-temporal sequence that embeds fine-grained human body information. A visual example is given in Fig. 1 (a), which illustrates the concept of *action classification*. In this setting, the input to the HAR algorithm is a fully-observed temporal sequence of human movements, hereby modeled by a skeleton graph. The HAR algorithm returns a set of class scores capturing the most likely action in the sequence. One of the prominent challenges of HAR is its efficient application to the continual setting, also referred to as *online action detection (OAD)*. This is visualized in Fig. 1 (b). In this case, inputs to the HAR algorithm are provided continually, which classifies them on a sample-by-sample basis [2] and based on the processing of previously observed samples. Thus, instead of a single prediction per fully-collected sequence the online system provides one prediction per received time sample.

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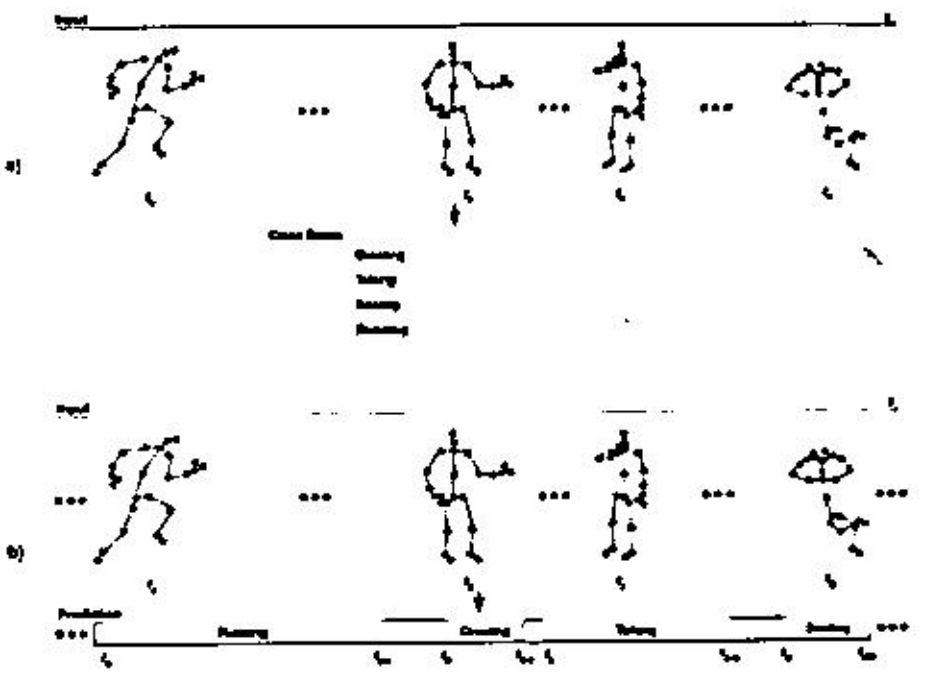

Fig. 1. (a) Action classification - one prediction per fully-observed temporal sequence. (b) Online detection - one prediction per observed time sample.

Continual HAR is essential for systems and applications in which predicting an action is time-critical, i.e. the time required for a prediction must be minimized in order to ensure a fast reaction time. If a system can provide a decision within a time *quant* (the time between reception of two consequent data samples) then it qualifies as *realtime* (RT). Time-critical and continual action monitoring has the potential to enable novel applications in several areas. In this work we are motivated by use cases in the healthcare and biomedical domains, in which our research group is active. Examples are systems for movement intent prediction, found in exoskeleton [3] and prosthesis control [4], which capture movement intent by using on-body sensors such as inertial measurement units (IMUs) or electromyography (EMG) sensors. In both settings, a realtime HAR implementation is necessary to continually drive the mechanical actuators in the exoskeleton and/or prosthesis to effectively assist the host's intended movement. Other use cases are found in systems for movement anomaly detection, like automatic detection of freezing-of-gait episodes in Parkinson's patients [5] that allows to provide sensory cues to aid the patient in overcoming the freezing event, or fall detection [6]